Analysis and Design of Algorithms

NUS National University of Singapore

CS3230

Probabilistic Analysis

- Basic probability theory
- Probabilistic analysis

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Axioms of Probability

Sample space *S*: a set whose elements are called **elementary events**.

- For throwing a dice, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$ consisting of the 6 possible outcomes which are the elementary events

An **event** is a subset of the sample space S.

- The event of throwing an even number is $A = \{2, 4, 6\}$

A probability distribution **Pr{}** is a mapping from events to real number satisfying

- $\Pr\{A\} \geq 0$ for any event A
- $\Pr\{S\} = 1$
- $\Pr\{A \cup B\} = \Pr\{A\} + \Pr\{B\}$ for any mutually exclusive events A and B

Example: A uniform distribution is a reasonable model for the throw of a fair dice

- $\Pr\{i\} = 1/6$ for $i = 1, \ldots, 6$
- If A is the set of even dice throws
  $\Pr\{A\} = \Pr\{2\} + \Pr\{4\} + \Pr\{6\} = 3/6$
  as they are mutually exclusive

- If A is the set of even dice throws and B is the set of dice throws whose outcome is greater than 3

  $\Pr\{A \cup B\} = \Pr\{A\} + \Pr\{B\} - \Pr\{A \cap B\}$

  $= \Pr\{2, 4, 6\} + \Pr\{4, 5, 6\} - \Pr\{4, 6\}$

  $= 3/6 + 3/6 - 2/6 = 4/6$

  A and B are not mutually exclusive in this case

Conditional Probability

- The **conditional probability** of an event A given another event B is defined as

$$\Pr\{A|B\} = \frac{\Pr\{A \cap B\}}{\Pr\{B\}}$$

whenever $\Pr\{B\} \neq 0$.

- Two events are **independent** if

$$\Pr\{A \cap B\} = \Pr\{A\}\Pr\{B\}$$

Example: Suppose we flip two fair coins. Let A_1 be the event that the first coin is heads, and A_2 be the event that the second coin is heads.

$$\Pr\{A_1\} = 1/2$$

$$\Pr\{A_2\} = 1/2$$

$$\Pr\{A_1 \cap A_2\} = 1/2 \times 1/2 = 1/4$$

Bayes's Theorem

- Bayes's theorem can be used for computing conditional probabities

$$\Pr\{A|B\} = \frac{\Pr\{A\}\Pr\{B|A\}}{\Pr\{B\}}$$

$$= \frac{\Pr\{A\}\Pr\{B|A\}}{\Pr\{A\}\Pr\{B|A\} + \Pr\{\bar{A}\}\Pr\{B|\bar{A}\}}$$

Example: Let A be the event of choosing a biased coin that always comes up head from a fair and biased coin. Let B be the event the the chosen coin comes up heads twice in two tosses. We have

$$\Pr\{A\} = 1/2, \Pr\{B|A\} = 1, \Pr\{\bar{A}\} = 1/2, \Pr\{B|\bar{A}\} = 1/4$$

$$\Pr\{A|B\} = \frac{(1/2)\cdot 1}{(1/2)\cdot 1 + (1/2)(1/4)} = 4/5$$

Random Variables

A random variable X is a function from the sample space S to the real numbers.

The function $f(x) = \Pr\{X = x\}$ is the probability density function of X.

Example: Rolling a pair of dice. Sample space contains 36 elementary events. Let X be *max* of the two values shown on the dice.

Then $\Pr\{X=3\}=5/36$: elementary events (1,3), (2,3), (3,3), (3,2), (3,1) has max of 3.

Expectation

The expectation or mean of a random variable X is

$$E[X] = \sum_x x \cdot \Pr\{X = x\}$$

Linearity of expectation

$$E[X + Y] = E[X] + E[Y]$$

$$E[aX] = aE[X]$$

If X and Y are independent

$$E[XY] = E[X]E[Y]$$

Geometric and Binomial Distributions

We can model a ***coin flip*** as an instance of a ***Bernoulli trial***, with probability p of success, and probability $q = 1 - p$ of failure.

Suppose we have a sequence of Bernoulli trials, each with prob p of success. Let X be the number of trials needed to obtain success. Then, X follows the ***geometric distribution***

$\Pr\{X = k\} = q^{k-1}p$ $\qquad$ $E[X] = 1/p$

Let X be the number of successes in n Bernoulli trials. Then X follows the ***binomial distribution***

$$\Pr\{X = k\} = \binom{n}{k} p^k q^{n-k} \qquad E[X] = np$$

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Probabilistic Analysis

Probabilistic analysis use probability in analyzing problems, usually the running time of algorithms

Need to know distribution of inputs – average the runtime over distribution of possible inputs. We refer to the runtime as *average-case running time*.

Randomized Algorithms

We call an algorithm **randomized** if its behaviour is determined not only by its input but also by values produced by a random-number generator.

We refer to the running time of a randomized algorithm as *expected running time* (distinguishing it from average-case running time of deterministic algorithms)

Indicator Random Variable

Indicator random variable

$$I\{A\} = \begin{cases} 1 & \text{if } A \text{ occurs} \\ 0 & \text{if } A \text{ does not occur} \end{cases}$$

Useful property:

Let $X_A = I\{A\}$. Then $\mathrm{E}[X_A] = \Pr\{A\}$

Example

Find the expected number of heads in n coin tosses.

Let X_i be the indicator random variable indicating that the i-th flip comes up head.

Let X be the total number of heads in n coin tosses.

$$X = \sum_{i=1}^{n} X_i$$

Sum of indicators

$$E[X] = E\left[\sum_{i=1}^{n} X_i\right]$$

Expected value

$$= \sum_{i=1}^{n} E[X_i]$$

Linearity of expectation

$$= \sum_{i=1}^{n} 1/2 = n/2.$$

$E[X_i] = \Pr\{X_i = 1\}$

Example: Hashing with Chaining

Given a hash table T with m slots that stores n elements, the load factor $\alpha = n/m$ is the average number of elements in a chain.

Assume that any given element is equally likely to hash into any of the m slots: ***simple uniform hashing***.

An unsuccessful search takes average-case time $\Theta(1 + \alpha)$ under assumption of simple uniform hashing.

- Expected number of elements in chain $h(k)$, $E[n_{h(k)}] = \alpha$ for key k and hash function h.
- Expected time to hash and search till end of chain is $\Theta(1 + \alpha)$.

Average-case time of successful search is also $\Theta(1 + \alpha)$.

- Assume elements inserted at the head of a list.
- Number of elements examined to find x is 1 plus the number of elements inserted after x.
- Let $X_{ij} = I\{h(k_i) = h(k_j)\}$.
- Under simple uniform hashing $\Pr\{h(k_i) = h(k_j)\} = \mathrm{E}[X_{ij}] = 1/m$.

$$E\left[\frac{1}{n}\sum_{i=1}^{n}\left(1+\sum_{j=i+1}^{n}X_{ij}\right)\right]$$

$$=\frac{1}{n}\sum_{i=1}^{n}\left(1+\sum_{j=i+1}^{n}E[X_{ij}]\right) \quad \text{(by linearity of expectation)}$$

$$=\frac{1}{n}\sum_{i=1}^{n}\left(1+\sum_{j=i+1}^{n}\frac{1}{m}\right)$$

$$=1+\frac{1}{nm}\sum_{i=1}^{n}(n-i)$$

$$=1+\frac{1}{nm}\left(\sum_{i=1}^{n}n-\sum_{i=1}^{n}i\right)$$

$$=1+\frac{1}{nm}\left(n^2-\frac{n(n+1)}{2}\right)$$

$$=1+\frac{n-1}{2m}$$

$$=1+\frac{\alpha}{2}-\frac{\alpha}{2n}$$

$$=\Theta(1+\alpha).$$

Coupon Collector's problem: There are n types of coupon that the collector would like to collect. The coupons are put into a box and randomly drawn with replacement. What is the expected number of times that the collector needs to draw from the box before he has at least one of each type of coupon?

Let T_i be the number of draws used to collect the i-th coupon and T be the total number of draws.

$$T = \sum_{i=1}^{n} T_i$$

Total number of draws

$$E[T] = E\left[\sum_{i=1}^{n} T_i\right]$$

Expected value

$$= \sum_{i=1}^{n} E[T_i]$$

Linearity of expectation

T_i is the time to collect the i-th coupon after i-1 coupon has been collected.

The prob of collecting a new coupon after i-1 coupon has been collected $p_i = (n-(i-1))/n$.

T_i has a geometric distribution with expectation $1/p_i = n/(n-(i-1))$.

$$
\begin{aligned}
E[T] &= \sum_{i=1}^{n} E[T_i] \\
&= \sum_{i=1}^{n} \frac{n}{n-(i-1)} \\
&= \frac{n}{n} + \frac{n}{n-1} + \cdots + \frac{n}{1} \\
&= n \cdot \left(\frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{n} \right) \\
&= n \cdot H_n \\
&= O(n \lg n)
\end{aligned}
$$

where H_n is the n-th harmonic number and is of size $O(\lg n)$

Summary

- Probabilistic analysis, usually of average-case runtime, requires knowledge of input distribution
- Randomized algorithm – algorithm uses its own randomization, often analyzed using expected runtime
- Linearity of expectation useful, often with indicator random variables, for analyzing average-case/expected runtime.